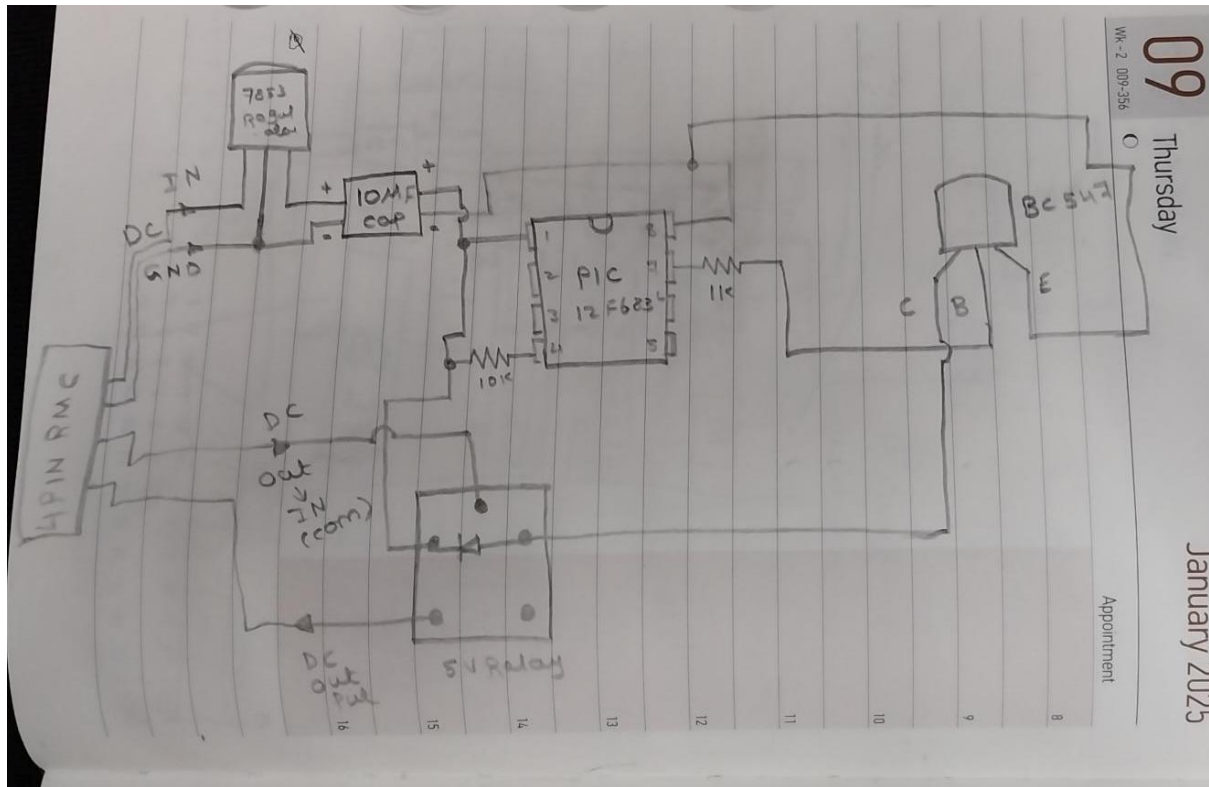


I-Tank Mini Timer Board:

1st Version (Old One – Change Time by Changing Program)



Source Code For I-Tank Mini Timer Board:

```
// PIC12F683 Configuration Bit Settings

#pragma config FOSC = INTOSCIO // Internal oscillator, GPIO on GP4/GP5

#pragma config WDTE = OFF // Watchdog Timer disabled

#pragma config PWRTE = ON // Power-up Timer enabled

#pragma config MCLRE = OFF // MCLR pin function is digital input

#pragma config CP = OFF // Code protection disabled

#pragma config CPD = OFF // Data code protection disabled

#pragma config BOREN = ON // Brown-out Reset enabled
```

```

#include <xc.h>          // Include header file for PIC12F683

#define _XTAL_FREQ 4000000 // Define the internal oscillator frequency (4 MHz)

void main(void) {

    // GPIO Initialization

    TRISIO = 0b11111000; // Set GP0 as output (relay control), others as input

    ANSEL = 0x00;        // Disable analog inputs (all pins are digital)

    CMCON0 = 0x07;        // Disable the comparator module

    GPIO = 0;             // Clear all GPIO pins initially


    while (1) {

        // Relay OFF for 3 hours

        GP0 = 0; // Set GP0 low (relay OFF)

        for (unsigned int i = 0; i < 10800; i++) { // 3 hours = 10800 seconds

            __delay_ms(1000); // Delay 1 second

        }


        // Relay ON for 1 minute

        GP0 = 1; // Set GP0 high (relay ON)

        for (unsigned int i = 0; i < 60; i++) { // 1 minute = 60 seconds

            __delay_ms(1000); // Delay 1 second

        }

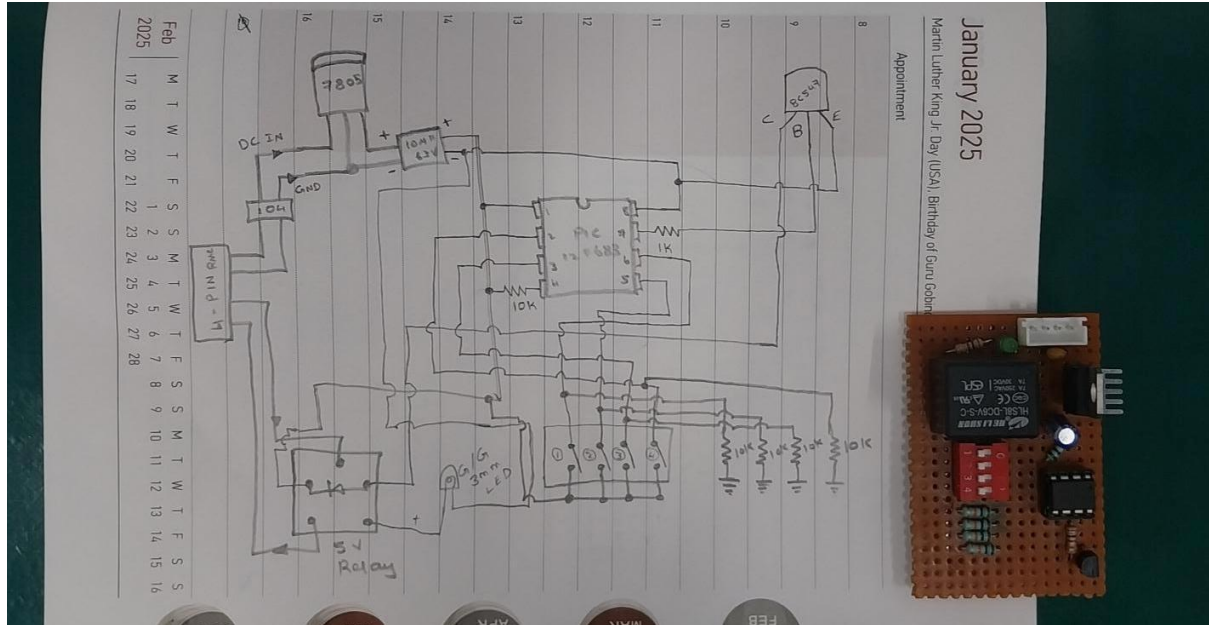
    }

}

```

I-Tank Timer V2 Board:

New Version 2 (Change Time By Using DIP Switch)



Source Code For I-Tank Timer V2 Board:

```
// PIC12F683 Configuration Bit Settings

#pragma config FOSC = INTOSCIO // Internal oscillator, GPIO on GP4/GP5

#pragma config WDTE = OFF // Watchdog Timer disabled

#pragma config PWRTE = ON // Power-up Timer enabled

#pragma config MCLRE = OFF // MCLR pin function is digital input

#pragma config CP = OFF // Code protection disabled

#pragma config CPD = OFF // Data code protection disabled

#pragma config BOREN = ON // Brown-out Reset enabled

#include <xc.h>

#define _XTAL_FREQ 4000000 // 4 MHz internal oscillator
```

```

// Define DIP switch pins

#define DIP1 GP1 // 1 Hour delay

#define DIP2 GP2 // 3 Hour delay

#define DIP3 GP4 // 8 Hour delay

#define DIP4 GP5 // 12 Hour delay


// Function to read DIP switch settings and determine off delay

unsigned int Get_Off_Delay() {

    unsigned int delay = 0;

    if (DIP1) delay += 3600; // 3600 sec

    if (DIP2) delay += 10800; // 10800 sec

    if (DIP3) delay += 28800; // 28800 sec

    if (DIP4) delay += 43200; // 43200 sec


    if (delay == 0) delay = 21600; // Default to 21600 sec(6 Hours) if no switch is ON

    return delay;

}


void Wait_Seconds(unsigned int sec) {

    while (sec--) {

        __delay_ms(1000); // 1 second delay

    }

}


void main(void) {

    // GPIO Initialization

    TRISIO = 0b11110110; // Set GP0 as output (relay), GP1, GP2, GP4, GP5 as inputs (DIP switches)

```

```
ANSEL = 0x00;    // Disable analog inputs
```

```
CMCON0 = 0x07;   // Disable comparators
```

```
GPIO = 0;        // Clear all GPIO pins
```

```
while (1) {
```

```
    unsigned int off_delay = Get_Off_Delay(); // Read DIP switch setting
```

```
    GP0 = 0; // Relay OFF
```

```
    Wait_Seconds(off_delay); // Wait based on selected off delay
```

```
    GP0 = 1; // Relay ON
```

```
    Wait_Seconds(60); // Fixed ON time of 60 seconds
```

```
}
```

```
}
```

DIP 0 – 6 Hours

DIP 1 – 1 Hour

DIP 2 – 3 Hour

DIP 3 – 8 Hours

DIP 4 – 12 Hours